



DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
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DEPARTMENT OF COMPUTER SCIENCE & BUSINESS SYSTEMS

UIUX LAB Manual

Semester IV

Course Code -22CBL473

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Program 1. Identifying interface connectivity and establishing interface connectivity between two different program modules

Problem statement: Designing Seamless Product Detail Navigation in an E-Commerce App

Objectives

- Understand how to design intuitive navigation flows in a mobile e-commerce app.
- Create interactive prototypes that simulate real app behavior using Figma.
- Ensure smooth and lag-free transitions between listing and product detail pages.
- Learn to apply smart transitions and interactive components for realistic UI simulation.

Topics Learnt

- UI layout design for product listing and detail views
- Interactive prototyping using Figma
- Smart Animate transitions and easing options
- Linking frames and managing user navigation flow
- Real-time testing and refinement of user interactions

Steps in Figma

1. Setup the Project

- Open Figma and create a new mobile frame (e.g., iPhone 13).

2. Design the Product Listing Page

- Add a grid or vertical list layout.
- Each product should include an image, name, and price.
- Group each product as a component for reusability.

3. Design the Product Detail Page

- Create a new frame for the detail view.
- Include:
 - Large product image
 - Product title and detailed description
 - Price and action buttons (e.g., Add to Cart)
 - Optional: ratings, size/color selectors, etc.

4. Create Interactions

- Switch to the **Prototype** tab.
- Link the product card (in the listing) to the product detail frame.

- **Trigger:** On Click
- **Action:** Navigate To
- **Animation:** Smart Animate or Instant (adjust easing as needed)

5. Add a Back Navigation

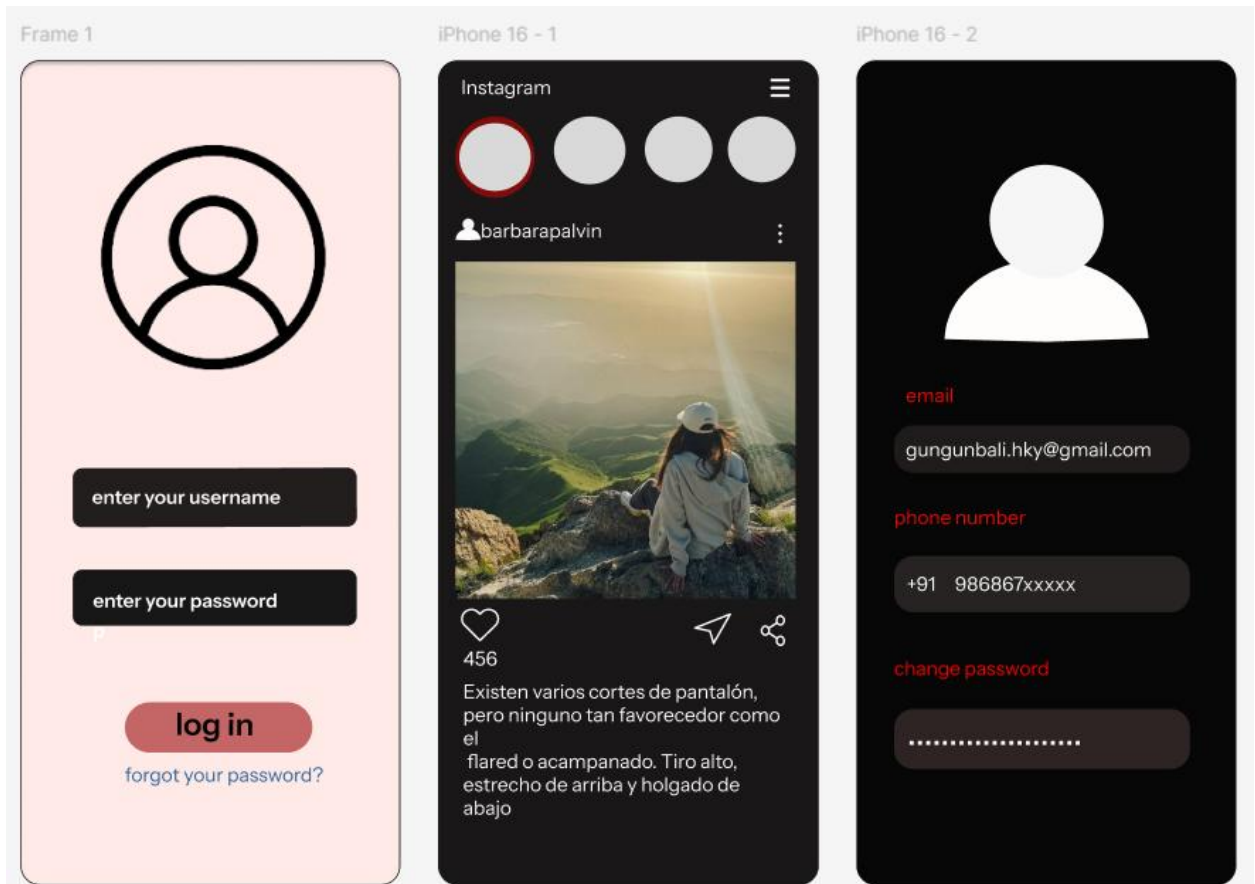
- Add a back button/icon on the detail screen.
- Link it back to the listing screen for two-way navigation.

6. Test the Prototype

- Click **Present** to test the interaction.
- Validate smoothness, transition logic, and clarity.
- Iterate based on issues found (e.g., image not animating due to mismatched names).

Expected Output

- A functional interactive prototype where:
 - Clicking a product card smoothly transitions to the product detail screen.
 - The user experiences intuitive and clear navigation with no confusion or lag.
 - UI components behave like a real app, demonstrating good UX practices.
- A ready-to-share prototype link suitable for usability testing or stakeholder demo.



Program 2: Understand front-end and back-end interfacing and implementation of both interfacing.

Problem statement: Designing Real-Time Feedback and User-Friendly Error Handling for Form Submissions

Objectives

- To learn how to simulate real-time feedback (e.g., processing indicators) in forms using Figma.
- Design user-friendly error messages that clearly guide users on how to resolve issues.
- Understand the importance of responsive UX during user interactions with forms.
- Apply interactive prototyping techniques to mimic dynamic UI states.

Topics Learnt

- Real-time visual feedback mechanisms
- Creating overlays or state changes for "Processing..." indicators
- Error messaging principles (clarity, tone, and guidance)
- Interactive transitions and timed delays in Figma prototypes

Steps in Figma

1. Design the Form UI

- Create a mobile frame with a form layout (e.g., email, password, submit button).
- Style input fields with proper labels and placeholders.
- Add a "Submit" button with clear visual prominence.

2. Design the "Processing..." Indicator Screen

- Duplicate the form screen or create a new one.
- Replace the button text with a **"Processing..."** message or add a loading spinner (can be simulated using ellipsis animation or circular shapes).
- Use a slight dimming overlay if needed to indicate inactivity.

3. Create Error State Screens

- Duplicate the form screen again.
- Simulate invalid input (e.g., leave a field blank or enter wrong email format).
- Add clear, polite error messages near the input field:
 - Example: *"Please enter a valid email address."*
 - Use red outlines or icons to signal the error visually.
- Add guiding hints like: *"Email should include '@' and domain"*

4. Prototype the Interactions

- Link the “Submit” button to the "Processing..." screen:
 - **Trigger:** On Click
 - **Action:** Navigate To
 - **Animation:** Instant or Smart Animate
- From the “Processing...” screen, use **After Delay (e.g., 1s)** interaction:
 - Link to either:
 - Success screen (optional)
 - Error state screen (to simulate failed input)

5. Add a Retry Option

- On the error screen, include a “**Try Again**” or “**Go Back**” button.
- Link it back to the form screen for correction.

6. Test the Prototype

- Click **Present**.
- Test the flow: form → processing → error → retry.
- Check that messages are readable, friendly, and helpful.

Expected Output

- A fully interactive prototype that:
 - Shows a “**Processing...**” state after clicking submit
 - Transitions to an **error state** with clear, actionable guidance
 - Allows the user to easily retry and fix the issue
- Demonstrates the importance of real-time feedback and empathetic error messaging in form UX.

iPhone 16 Pro Max - 1

iPhone 16 Pro Max - 2

iPhone 16 Pro Max - 3

iPhone 16 Pro Max - 4

iPhone 16 Pro Max - 5

Program 3. Identifying interaction design and functional layout. Practical implementation of interaction design and functional layout.

Problem statement: Designing Intuitive Registration Form for Marathon Event with Interaction Design and Functional Layout Principles

Objectives

- Understand the principles of interaction design and functional layout in UI/UX.
- Apply logical structuring to form fields for better usability.
- Design clear visual cues and user-friendly interactions to guide users through form filling.
- Simulate real-time feedback and error handling in forms to enhance user engagement.

Topics Learnt

- Functional layout: alignment, grouping, and spacing for clarity
- Form usability: label placement, field order, and error prevention
- Micro-interactions for feedback (e.g., validation, hover/focus effects)
- Building a user-centric form experience in Figma

Steps in Figma

1. Plan the Form Structure

- Determine what information the form needs (e.g., name, age, address and so on).
- Decide the logical order (e.g., personal info → credentials → action button).

2. Design the Base Form Layout

- Create a mobile or desktop frame in Figma.
- Use vertical alignment with proper spacing (8–16px between fields).
- Place labels **above** input fields for better readability.
- Add placeholder text if needed (e.g., “Enter your email”).

3. Group Related Elements

- Use visual grouping for related fields (e.g., address lines).
- Maintain consistent padding and alignment using Figma's layout grids or auto-layout.

4. Add Action Button

- Include a clear **Submit** or **Sign Up** button below the fields.
- Style it to stand out (e.g., primary color, full-width, large enough for touch).

5. Design Interaction Feedback

- On field focus: show a highlight or border color change.

- On submit: simulate feedback like “Processing...” or form validation.

6. Prototype the Interaction Flow

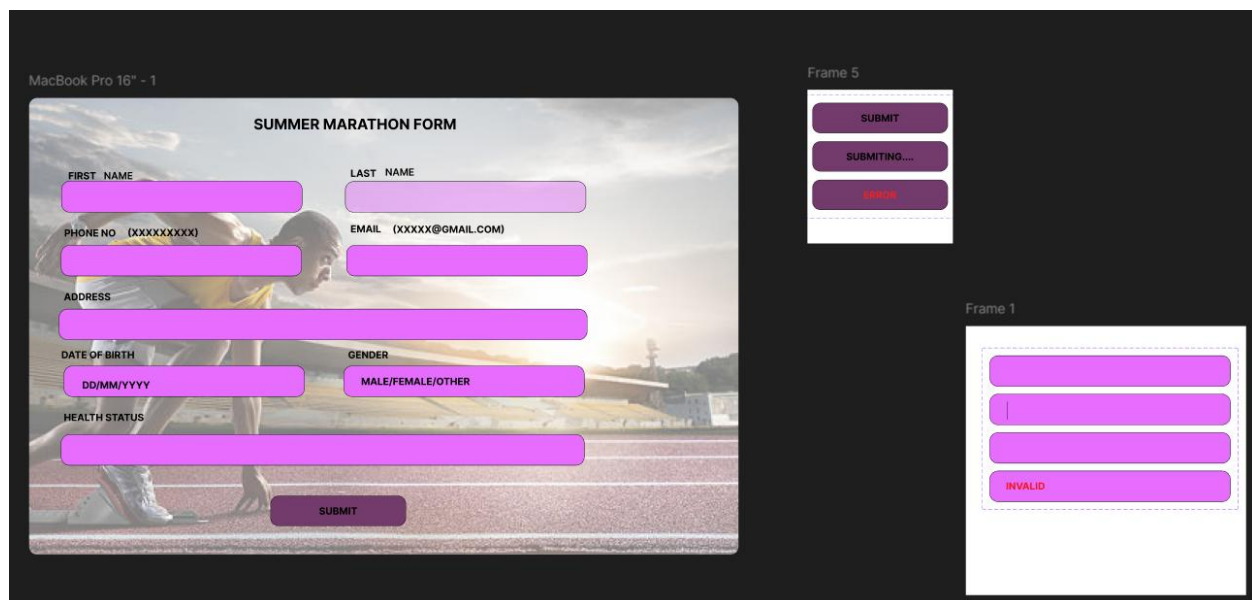
- Use the **Prototype** tab to connect:
 - Form → Processing screen (with delay)
 - Processing → Success or Error state
- Add hover/focus/fill states using **Variants** or **Interactive Components**.

7. Test and Iterate

- Use **Present** mode to test the form interaction.
- Validate clarity, ease of use, and feedback visibility.
- Refine based on flow and visual hierarchy.

Expected Output

- A well-structured form prototype with:
 - Logically placed fields and clear labels
 - Real-time interaction feedback (e.g., field focus, validation)
 - User-friendly error messages guiding next steps
 - Seamless flow from form entry to submission and response
- An intuitive interface that reflects strong interaction design and functional layout principles





Marathon Application Form

Full name

First Last

Email address

EMAIL

Phone no

Ph no

Birth day

DD/MM/YYYY

Program 4: Identify and analyze “what is navigation design” and implementing of navigation design.

Title: Designing Effective Navigation Systems in Web and Mobile Interfaces using top navigation

Objectives

- Understand the importance of intuitive navigation in digital applications.
- Explore different types of navigation patterns and when to use them.
- Design and prototype navigation components in Figma for both web and mobile contexts.
- Simulate user movement through an app using various navigation styles (e.g., sidebar, hamburger menu, breadcrumbs).

Topics Learnt

- Navigation design principles: clarity, consistency, hierarchy, and discoverability
- Types of navigation:
 - **Top Navigation** (horizontal menu)
 - **Sidebar Navigation** (vertical menu for dashboards)
 - **Breadcrumbs** (location context)
 - **Hamburger Menus** (mobile-friendly)
- Interactive components for navigation
- Linking and prototyping multi-screen flows in Figma

Steps in Figma

1. Choose Your App Type

- Decide on the application context:
 - Website (top nav or breadcrumbs)
 - Dashboard (sidebar)
 - Mobile app (hamburger menu)

2. Create the Main Screens

- Design at least 3–4 different screens (e.g., Home, Products, Profile, Settings).
- Use consistent layout and spacing across screens.

3. Design Navigation Components

- **Top Navigation:** Place a horizontal bar at the top with links to major sections.
- **Sidebar Navigation:** Design a vertical menu on the left with icons and labels.

- **Breadcrumbs:** Add a trail above the content area showing the user's path.
- **Hamburger Menu:**
 - Create a menu icon ()
 - Design a slide-out panel for links (can use an overlay or side transition)

4. Add Navigation Logic with Prototyping

- Use the **Prototype** tab to link nav items to respective screens.
- For hamburger menus:
 - Use **Overlays** or a new frame that slides in
 - Set trigger as **On Click**
- For breadcrumbs:
 - Link each segment to its corresponding screen

5. Add Active State Indicators

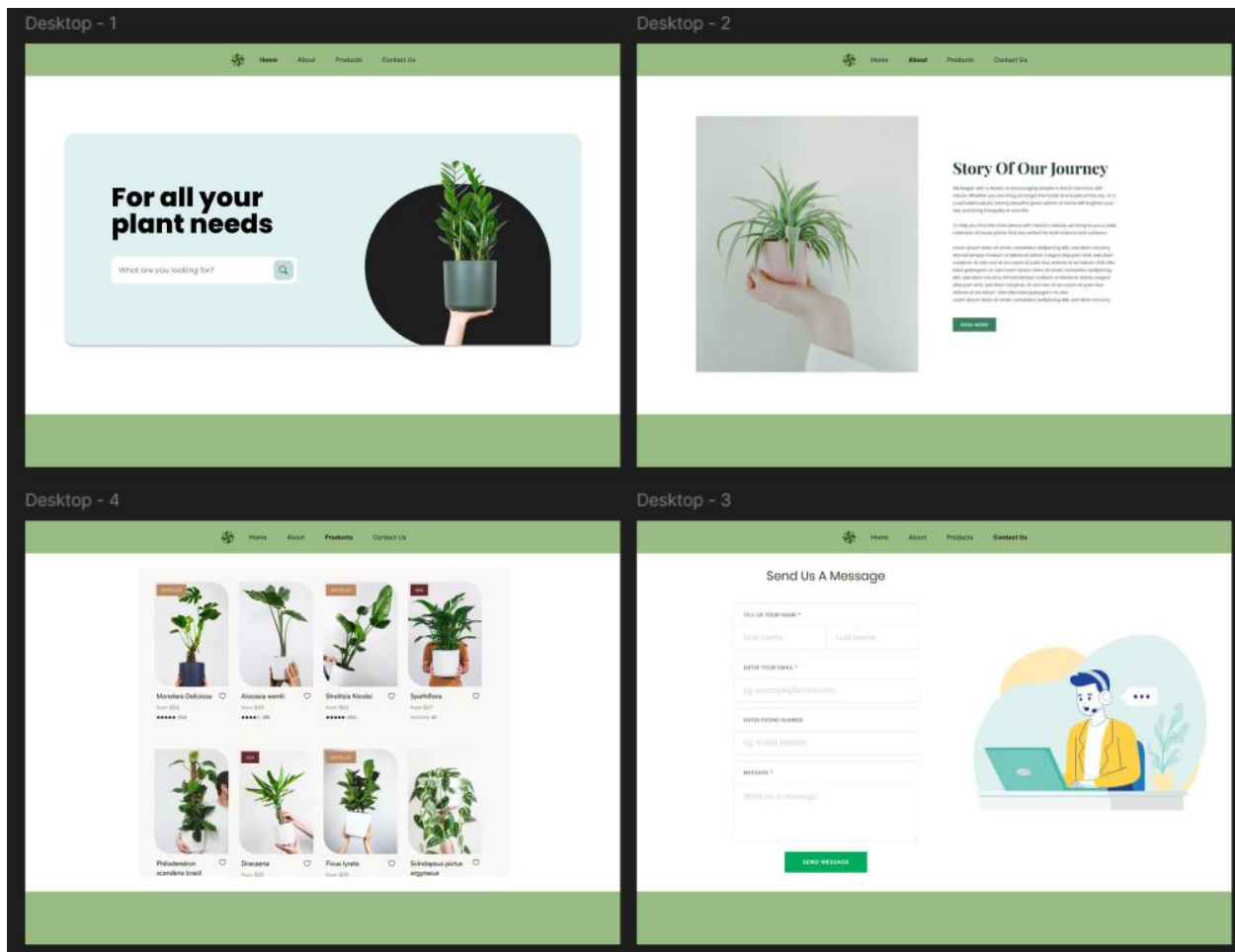
- Highlight the current section (e.g., bold text, underline, or background color) to show where the user is.
- Use **Variants** for reusable nav components with different states (active, hover, default).

6. Test Navigation Flow

- Click **Present** to walk through the screens.
- Check if users can easily move between sections and understand where they are.

Expected Output

- A multi-screen interactive prototype with:
 - Functional and user-friendly navigation (top, sidebar, breadcrumbs, or hamburger)
 - Clearly labeled links or icons
 - Active states that show the user's current location
 - Smooth transitions between screens
- A navigation system that reflects clarity, usability, and contextual awareness



Program 5: Create a working UI/UX prototype using prototyping tools.

Problem statement: A prototype of a mobile banking app, allowing users to click through the deposit and transfer flow.

Objectives

- Understand the purpose and benefits of prototyping in the design process.
- Learn how to build interactive flows using Figma's prototyping tools.
- Simulate real user behavior to test and validate UI/UX decisions before development.
- Create a clickable prototype of a core app feature (e.g., deposit or transfer in a banking app).

Topics Learnt

- What is prototyping and why it matters in UX/UI design
- Differences between low-fidelity and high-fidelity prototypes
- Using Figma's **Prototype** tab to create screen transitions and user flows
- Adding interactions, overlays, smart animations, and delays
- Usability testing basics with interactive prototypes

Steps in Figma

1. Define the User Flow

- Choose a simple flow to prototype (e.g., in a mobile banking app):
 - Home → Deposit → Enter Amount → Confirm
 - OR Home → Transfer → Select Account → Enter Amount → Confirm

2. Design the UI Screens

- Create 3–5 mobile frames to represent each step in the user journey.
- Add consistent headers, icons, inputs, and buttons.
- Keep layout clean and focused on the core task (e.g., transferring funds).

3. Switch to Prototype Mode

- Go to the **Prototype** tab in the right panel.
- Select tappable components (e.g., buttons) and **drag connectors** to the target screen.

4. Set Interactions and Transitions

- Choose interaction types:
 - **On Tap / Click → Navigate To → [Target Frame]**
 - Add **Smart Animate** or **Instant** transitions for realism.

- Optional: Use **After Delay** to simulate loading screens or success messages.

5. Add Overlay Screens (Optional)

- Design a modal for confirmation or a bottom sheet for account selection.
- Use **Overlay** interaction to display it without leaving the current screen.

6. Test and Refine

- Click **Present** to test the prototype.
- Walk through the full flow:
 - Tap buttons → navigate → see feedback → reach success state.
- Identify any UI/UX issues or breakpoints and refine the design.

Expected Output

- A functional interactive prototype of a mobile banking flow:
 - Smooth, logical navigation between steps (deposit or transfer)
 - Clearly visible call-to-action buttons and feedback messages
 - Use of overlays, transitions, or delays where appropriate
- A shareable prototype link that can be used for **user testing**, **peer review**, or **stakeholder presentations**

